

Ridge and Lasso Regression

Regularized Linear Models

राष्ट्रीय इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी संस्थान

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L2 ball



L1 ball

The roadmap

#	Section	Focus	Min
1	Why regularization?	Overfitting, huge coefficients	8
2	Recap: linear regression & MSE	What the loss actually measures	6
3	Ridge (L2)	Penalty, shrinkage, alpha	10
4	Lasso (L1)	Exact zeros, feature selection	10
5	Ridge vs Lasso	Comparison table, the geometry	8
6	Scaling & pipelines	Why order of operations matters	6
7	California Housing demo	Results, tuning alpha, CV	12
8	Wrap-up + quiz	Cheat sheet, questions	5
	Total		60

Timings are a guide — sections 3 and 4 are the core; trim section 7 if the room needs more time on the basics.

Why plain linear regression overfits

- Few rows, many features — the model has enough freedom to memorise the training set.
- Nothing in the loss function stops the coefficients from growing.
- A tiny change in one feature then swings the prediction wildly.

THE FIX, IN ONE LINE

$$\text{Loss} = \text{MSE} + \alpha \times \text{Penalty}$$

Charge the model for the size of its coefficients.

HOUSE PRICES · 8 ROWS, 12 FEATURES

Size (sq ft)	+48.2
Bedrooms	-37.9
Distance to city	+51.4
Garage spaces	-44.6
Training MSE	0.00
Test MSE	41.70

Perfect on the data it has seen. Useless on the next house.

THINK ABOUT IT · 60 SECONDS

What happens if a model gives huge coefficients to reduce training error?

$$\beta_1 = 812.4$$

$$\beta_2 = -903.7$$

$$\beta_3 = 764.1$$

Discuss with the person next to you before we move on.

Linear regression in one line

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

Prediction = a starting value, plus each feature multiplied by its own weight.

 $\hat{y}$

Prediction

what the model outputs for one row

 $\mathbf{x}$

Feature

one measured input — size, income, age

 β

Coefficient

how strongly that feature moves the prediction

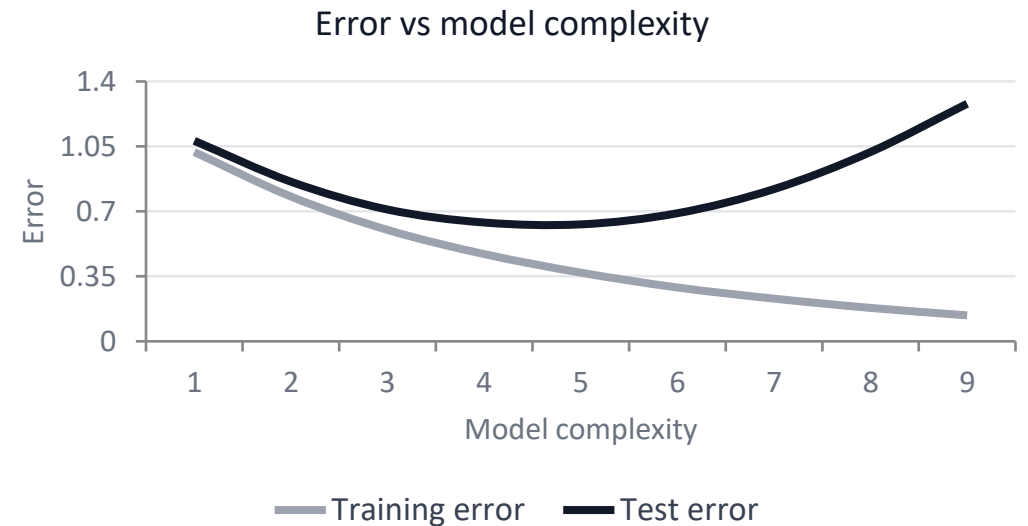
What the model is minimising

$$\text{MSE} = 1/n \sum (y_i - \hat{y}_i)^2$$

On average, how far off are the predictions — squared, so big misses hurt more.

Low training error $\neq$ good generalization

- Least squares only ever sees the training rows.
- Push complexity far enough and training error keeps falling — while test error turns around and climbs.
- The gap between the two curves is what we are trying to close.



Regularization: add a price tag to complexity

$$\text{Loss} = \text{MSE} + \alpha \times \text{Penalty}$$

Fit the data well and keep the coefficients small — the model must now do both.

Ridge

L2

$$\alpha \sum \beta_j^2$$

Squares the coefficients. Shrinks them all.

Lasso

L1

$$\alpha \sum |\beta_j|$$

Uses absolute values. Can zero them out.

Alpha: how hard the penalty pushes

$\alpha = 0$



No penalty at all.
You are back to plain
linear regression.

Small α



Gentle shrinkage.
Coefficients move a little
toward zero.

Large α



Strong shrinkage.
Coefficients collapse —
the model underfits.

Alpha is not a coefficient

The model learns β from the data. You choose α before training — **it is a hyperparameter**, like k in k -NN. Bigger α does not mean a better model; it means a simpler one. The right value is found by **cross-validation**, not by taste.

Ridge regression

$$\text{Loss} = \text{MSE} + \alpha \sum \beta_j^2$$

Fit the data and pay a price equal to the sum of the squared coefficients.

- **Squared penalty**

Doubling a coefficient quadruples its cost, so the biggest coefficients are punished hardest.

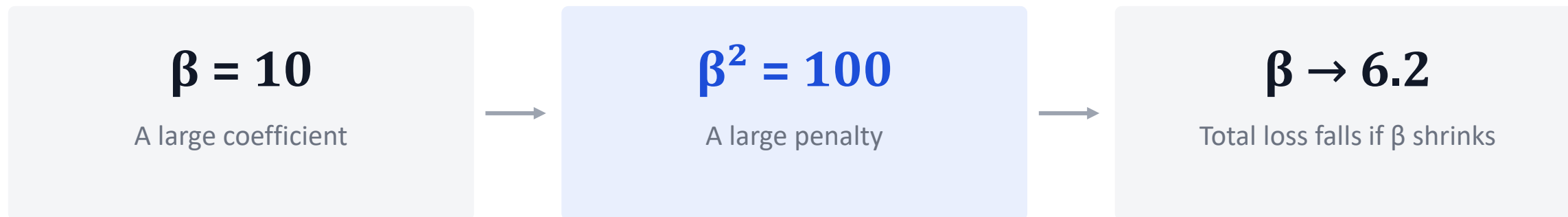
- **Everyone shrinks**

All coefficients move toward zero together — none of them is ever dropped.

- **Keeps all features**

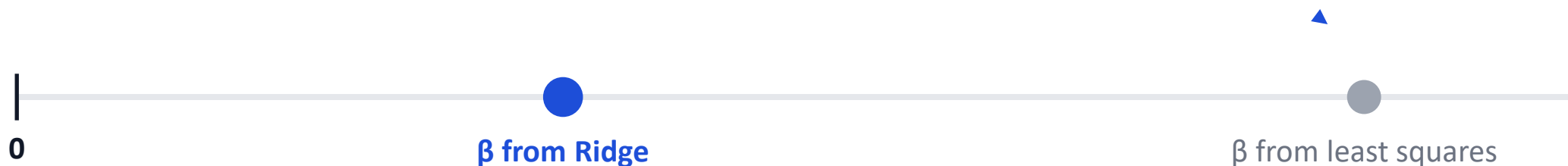
Coefficients get small but stay non-zero. Ridge never performs feature selection.

Why squaring shrinks the big coefficients



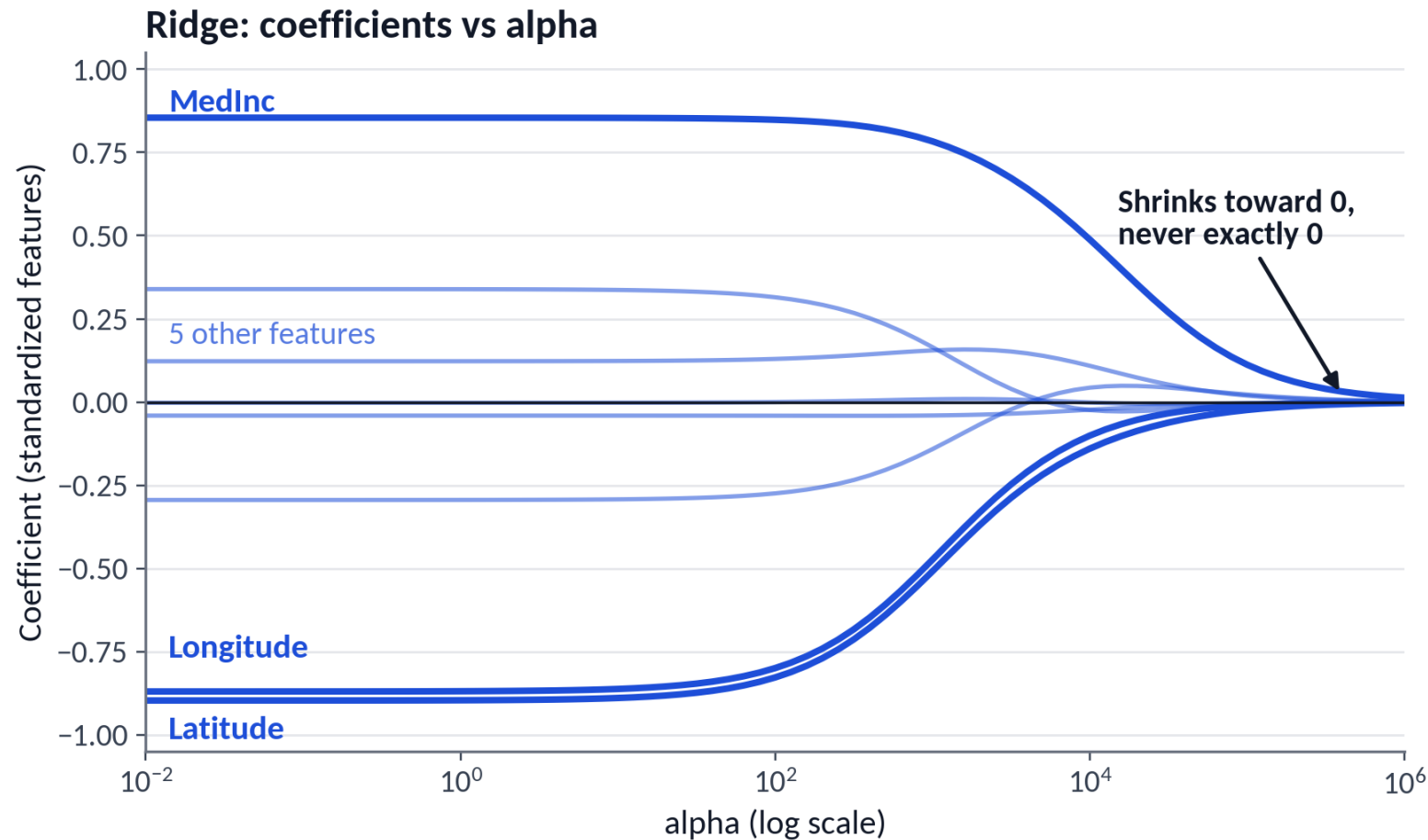
SHRINKAGE, SEEN ON THE NUMBER LINE

Shrinks toward zero — and stops short of it



As α grows the blue dot slides left, but it only reaches zero in the limit — never at any finite α .

Ridge coefficients as alpha grows



From the notebook

- Every curve is one feature's coefficient.
- Small α : identical to plain linear regression.
- Large α : all curves bend toward the zero line together.
- None of them ever touches it.

Shrinkage, not selection.

Lasso regression

$$\text{Loss} = \text{MSE} + \alpha \sum |\beta_j|$$

Fit the data and pay a price equal to the sum of the coefficient sizes, ignoring sign.

- **Absolute values**

Cost grows at a constant rate, so the pressure to shrink never fades as β approaches zero.

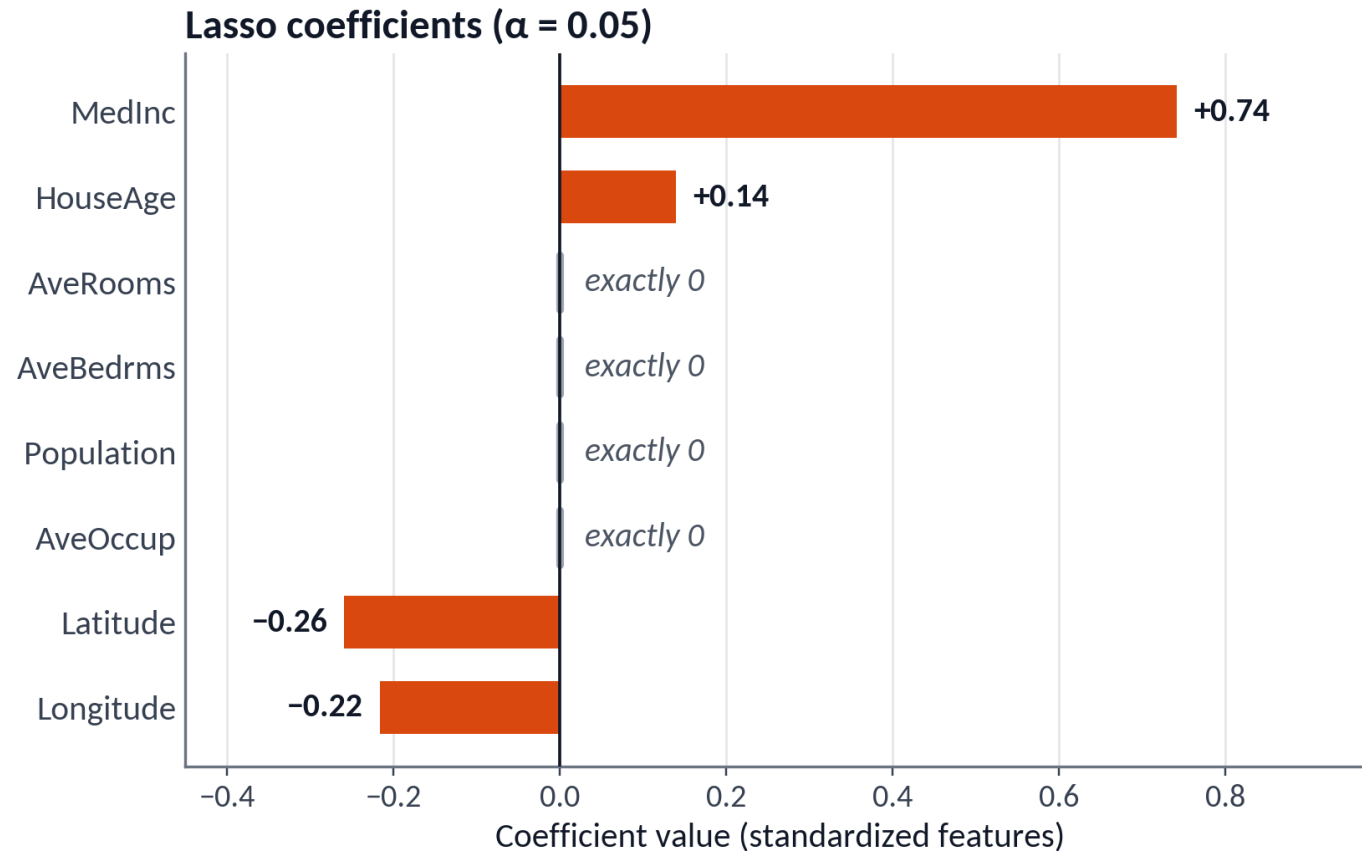
- **Exact zeros**

Coefficients can be pushed all the way to zero and stay there.

- **Feature selection**

A zero coefficient means the feature is out of the model — Lasso chooses for you.

Lasso switches features off



From the notebook

- Four of the eight coefficients are exactly 0.00 — not small, zero.
- Those four features are gone: they contribute nothing to any prediction.
- MedInc survives with the largest weight, and location keeps a smaller say.
- Raise α further and even MedInc eventually goes.

Shrinkage and selection.

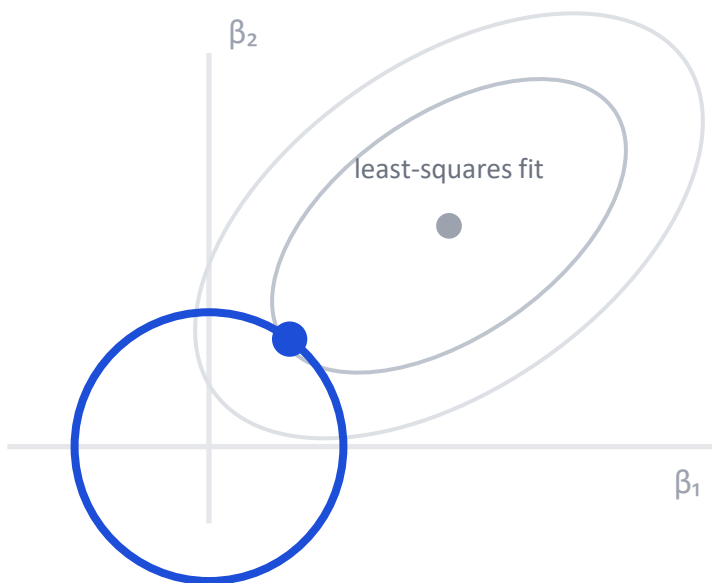
Ridge vs Lasso

Property	Ridge	Lasso
Regularization type	L2	L1
Penalty term	$\alpha \sum \beta^2$	$\alpha \sum \beta $
Shrinks coefficients	Yes	Yes
Coefficients exactly zero	Never	Yes
Feature selection	No	Built in
Reduces overfitting	Yes	Yes
Solution	Closed form, fast	Iterative (coordinate descent)
Correlated features	Shares weight between them	Keeps one, drops the rest

Same MSE term, same alpha dial. Only the shape of the penalty differs — and that one difference produces every row above.

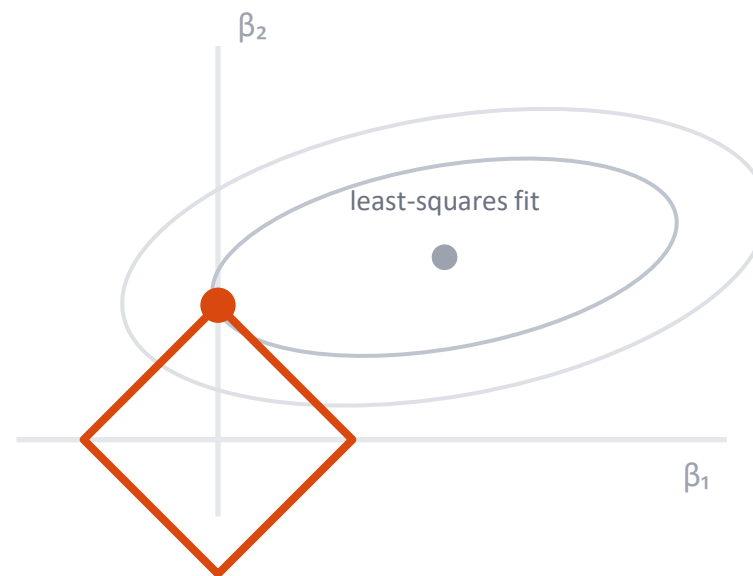
Why Lasso lands on zero: circle vs diamond

Ridge • the L2 region is a circle



The contours meet the circle on a smooth edge, so both coefficients stay non-zero.

Lasso • the L1 region is a diamond



The contours meet the diamond at a corner — and a corner sits on an axis, where $\beta_1 = 0$.

The shape has corners on the axes. That is the entire reason Lasso produces exact zeros.

Scale first — or the penalty is unfair

Age

18 – 65

Experience

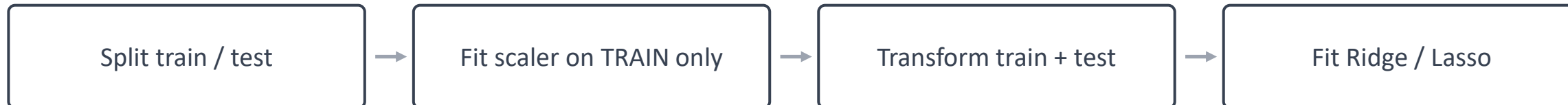
0 – 40

Salary

20,000 – 200,000

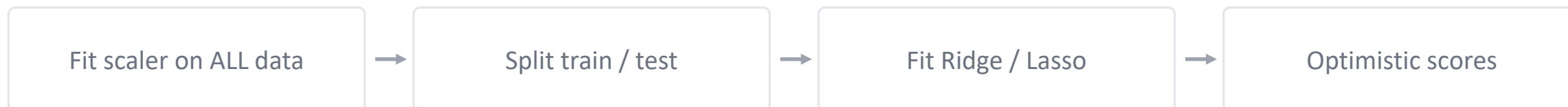
The penalty only sees the numbers. Salary's coefficient is tiny because its units are huge, so it escapes the penalty while Age is crushed — a difference in units becomes a difference in regularization.

CORRECT ORDER



The test set is scaled using the training set's mean and standard deviation.

WRONG ORDER — DATA LEAKAGE



The scaler has already seen the test rows, so your evaluation is quietly inflated.

One object that does it in the right order

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge

model = make_pipeline(StandardScaler(), Ridge(alpha=1.0))
model.fit(X_train, y_train)
```

● Order is guaranteed

Scale, then fit — every time, in that sequence.

● No leakage in CV

Inside `cross_val_score` the scaler is refitted on each training fold.

● One object to tune

`GridSearchCV` can search over the `alpha` inside the pipeline.

California Housing: does it actually help?

Model	MAE	RMSE	R ²	Features used
Linear Regression	0.534	0.746	0.576	8 of 8
Ridge ($\alpha = 1.0$)	0.534	0.746	0.576	8 of 8
Lasso ($\alpha = 0.01$)	0.535	0.740	0.582	7 of 8
Lasso ($\alpha = 0.05$)	0.582	0.784	0.531	4 of 8

20,640 rows · 8 features · 80 / 20 split, random_state = 42 · features standardised · target in \$100k

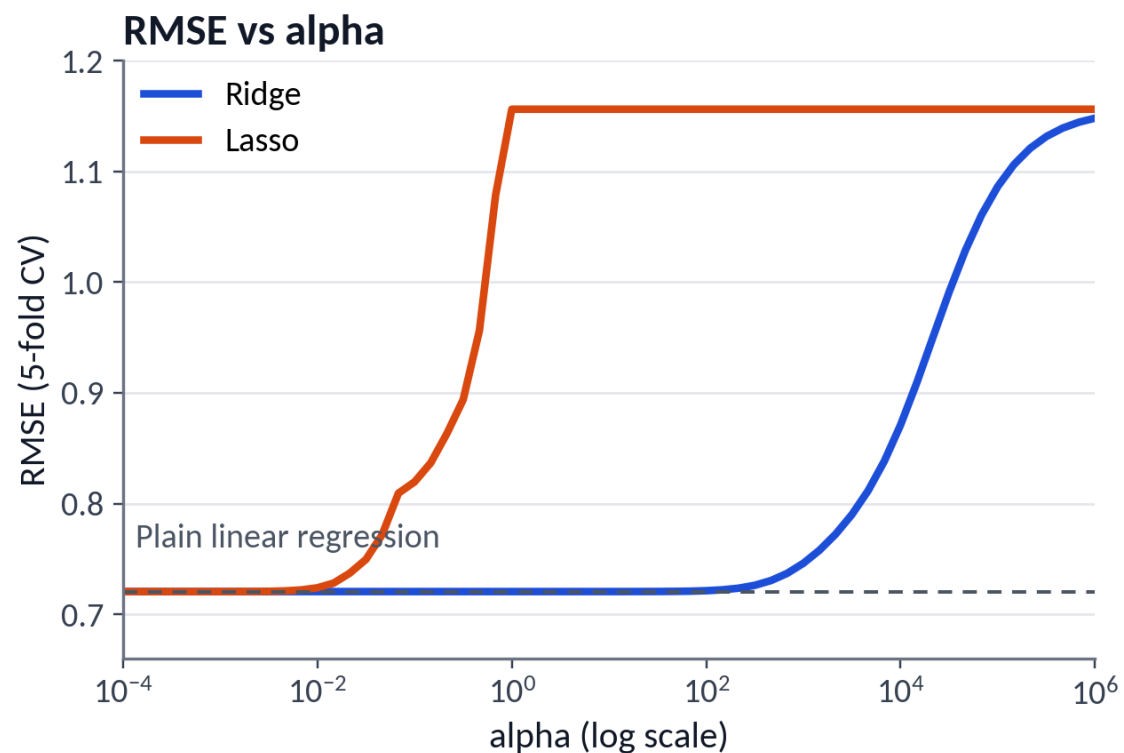
Accuracy barely moves

With 16,512 training rows and only 8 features, plain regression was never overfitting much — there is little for the penalty to fix.

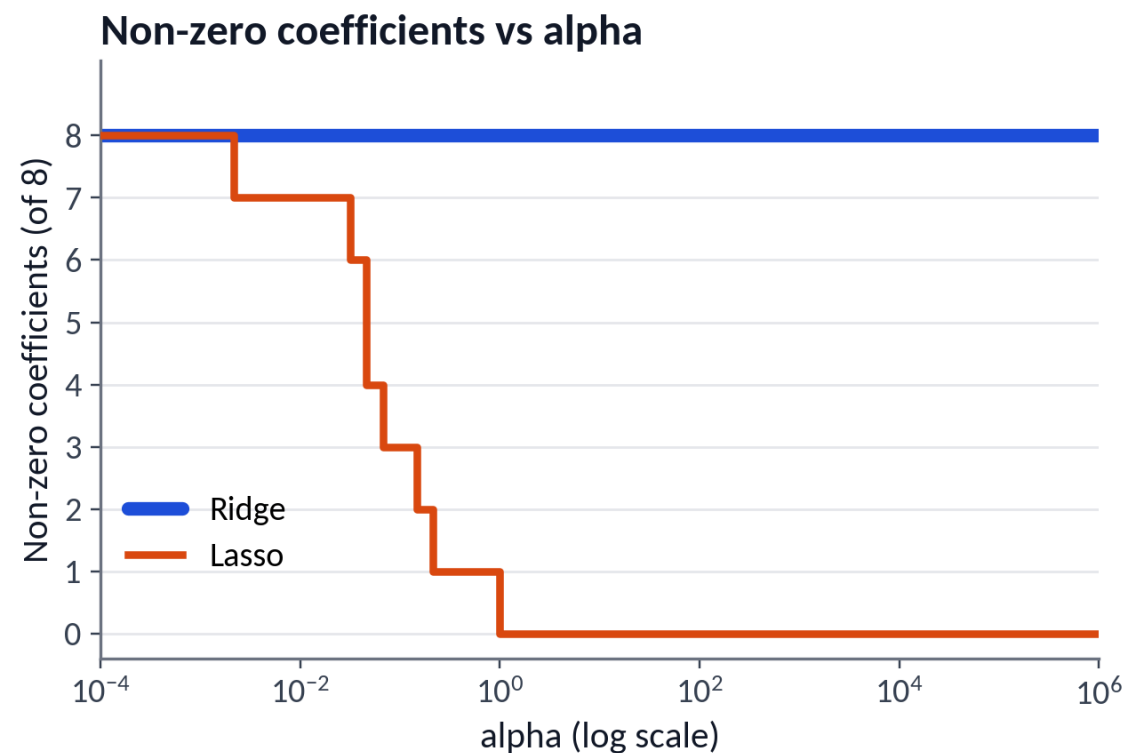
But the model gets smaller

At $\alpha = 0.05$ Lasso keeps half the features for a modest cost in RMSE. That trade is the real product here.

Tuning alpha: the two curves to watch



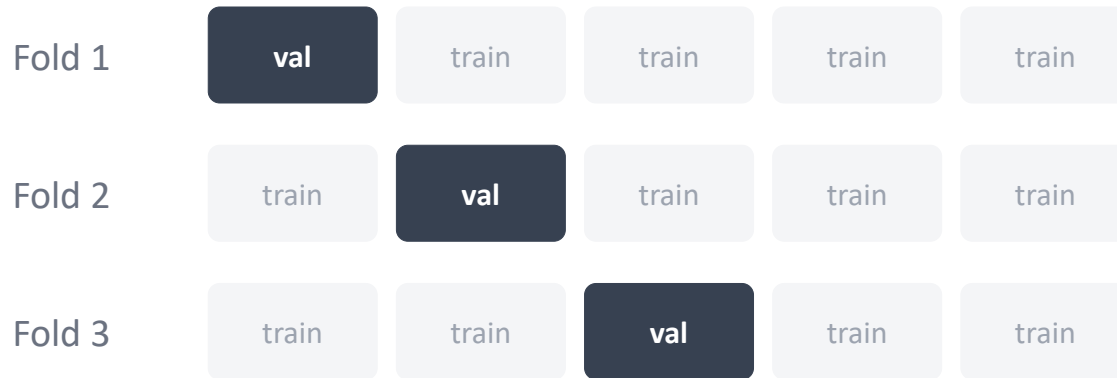
Flat, then a cliff — Ridge tolerates a far wider range of α .



Ridge keeps all 8 features; Lasso steps down to zero.

Choose alpha by cross-validation

5-FOLD CROSS-VALIDATION ON THE TRAINING SET



...and so on for folds 4 and 5. Average the five scores — that is the score for one value of α .

```
RidgeCV(alphas=np.logspace(-3, 3, 50), cv=5)
LassoCV(cv=5, random_state=42)
```

- Split the training data into 5 parts; train on 4, score on 1, rotate.
- Repeat for every candidate α and keep the best average score.
- RidgeCV and LassoCV do exactly this in one call — and they are faster than a manual loop.

The test set is for the final evaluation only

Touch it once, at the end. Tune α there and your reported score stops being an estimate of anything.

Which one should you reach for?

Use Ridge when

- Most features carry some signal and you want to keep them all.
- Features are correlated — Ridge shares weight instead of picking arbitrarily.
- You want a stable, fast, well-behaved model.

Use Lasso when

- You suspect many features are irrelevant.
- You want an automatic shortlist of what matters.
- A smaller, cheaper, more explainable model is worth a little accuracy.

Careful with correlated features

Lasso keeps one of two near-identical predictors and zeroes the other — and the choice can flip on a different sample. It is not evidence of importance.

Elastic Net

Both penalties at once: selects features like Lasso, keeps correlated groups together like Ridge. The usual answer when you cannot decide.

Six mistakes to avoid

1

Not scaling the features

The penalty then measures units, not importance.

3

Treating α as something the model learns

You choose it before training — it is not in `.coef_`.

5

Expecting Ridge to give exact zeros

It shrinks toward zero and stops short. Only Lasso zeroes.

2

Fitting the scaler before the split

Test statistics leak into training. Use a Pipeline.

4

Tuning α on the test set

Cross-validate on the training data instead.

6

Reading Lasso's zeros as importance

With correlated features the choice is close to arbitrary.

Four questions before we close

1

If $\alpha = 0$, what model are you actually fitting?

2

Which method can set a coefficient to exactly zero — and why?

3

Why must features be standardised before regularizing?

4

You tried 20 values of α on the test set. What is wrong with your reported score?

Cheat sheet

Ridge

$$L2 \cdot \alpha \sum \beta^2$$

- Penalty: sum of squared coefficients
- Shrinks every coefficient toward zero
- Never reaches exactly zero
- Keeps all features, shares weight across correlated ones
- Closed-form solution, fast

Lasso

$$L1 \cdot \alpha \sum |\beta|$$

- Penalty: sum of absolute coefficients
- Shrinks, and can set coefficients to exactly zero
- Performs automatic feature selection
- Among correlated features, keeps one
- Iterative solver, slightly slower

True for both

- Standardise the features first
- $\alpha = 0$ gives plain linear regression
- α is chosen by you, tuned by cross-validation
- Trade a little bias for a lot less variance

- Ridge controls coefficient magnitude;
- ◇ Lasso can control coefficient magnitude and eliminate features.

Minute-by-minute flow

Clock	What happens	Slides	Note to self
00 – 03	Title, roadmap	1 – 2	Set the colour convention: blue Ridge, orange Lasso
03 – 11	Overfitting, discussion prompt	3 – 4	Give the full 60 seconds on slide 4
11 – 17	Linear regression and MSE recap	5 – 6	Move fast; most of the room knows this
17 – 22	Loss = MSE + penalty, alpha	7 – 8	The two most important slides of the hour
22 – 32	Ridge: formula, shrinkage, chart	9 – 11	Land: shrinks but never zero
32 – 42	Lasso: formula, bar chart, table	12 – 14	Land: exact zeros, feature selection
42 – 46	Geometry, scaling, pipeline	15 – 17	Keep geometry intuitive — no optimisation theory
46 – 55	California Housing, tuning, CV	18 – 20	Switch to Colab here if running live
55 – 60	Use cases, mistakes, quiz, close	21 – 25	Quiz first; cheat sheet can be self-study

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